Short-Term Runway Wind Vector Prediction & Component Analysis via XGBoo

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EXECUTIVE ABSTRACT & SYSTEM SCOPE

Individual thesis focused on hyper-local surface wind speed and direction forecasting for runway 04/22 at Bandaranaike International Airport. Combines gradient boosted trees and Temporal Fusion Transformers to generate operational headwind and crosswind components.

1. Operational Significance of Runway Wind Analysis

Crosswind limits are decisive safety thresholds for commercial airliner landing operations. Sudden gusts during monsoon transitions cause unstable approaches and go-arounds at VCBI.

2. Feature Engineering & Multi-Horizon Modeling

Continuous wind direction is encoded using sine and cosine cyclical transformations to overcome the 0/360 degree discontinuity.

Models tested: XGBoost regression on lag features vs Temporal Fusion Transformer (TFT) utilizing multi-head self-attention mechanisms for multi-horizon forecasts.

3. Results and Runway Component Integration

" XGBoost Baseline MAE: 1.69 knots

" Temporal Fusion Transformer (TFT) MAE: < 1.0 knot (superior temporal attention)

" Automated headwind and crosswind decomposition provides real-time runway occupancy and safety advisories within the interactive pilot portal.